Meaning Between Lenses:

The Action-Theoretic Turn in Semantic Evolution

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Abstract

We propose that inter-lens meaning is the divergence between the act a speaker attempts and the act a listener experiences under culturally constituted interpretive lenses. We introduce a measurable framework—mapping strings to distributions over interactional acts—and show how recurrent indirect-speech patterns across genealogically independent cultural settings instantiate a common structural signature under face, cooperation, and hierarchy pressures. Drawing on convergent evolution evidence from eight cultural systems spanning five unrelated language families (English-family, Japanese, Korean, Persian, and Akan), we demonstrate that the structural isomorphism of indirect communication is statistically confirmed and unlikely to arise by chance. Hofstede's Power Distance Index correlates positively with indirect feedback frequency across 13 cultures (Spearman $\rho = .69$, $p = .009$) in an exploratory analysis supporting the model's hierarchical ambiguity component. We formalize decoding conditions in a Cultural–Relational Lens Matrix (CRLM), which identifies two independent lens sources, four decoding conditions, mirror-image inversion patterns, and two systematic failure modes (Underdecode/Overdecode). The framework recovers Gricean implicature, Relevance Theory, and strategic indirect speech as special cases under specific boundary conditions, and generates five testable predictions for corpus and experimental validation, including time-dependent retroactive re-interpretation after lens updates. We argue that cross-cultural miscommunication is not communicative failure but the condition under which inter-lens meaning becomes observable.

Keywords: *meaning, indirect speech, convergent evolution, cultural lenses, action theory, cross-cultural communication, pragmatics, language evolution*

1. Introduction

This paper treats meaning as an interactional observable in evolutionary pragmatics: not a metaphysical locus, but a measurable divergence in act inference across interpretive lenses.

Where is meaning? For over a century, this question has organized inquiry in philosophy of language, linguistics, and cognitive science. Saussure (1916) located meaning in the sign itself. Grice (1975) shifted the locus to speaker intention. Sperber and Wilson (1986) reformulated this as cognitive relevance. In each case, meaning is treated as a noun—a static object with a determinate location. This paper argues that the productive question is not *where* meaning is, but *what meaning is doing*. The shift from ontology to action theory transforms meaning from a noun into a verb—a dynamic process occurring between participants. This connects directly to evolutionary inquiry, where functional questions ("What is this trait doing?") are the natural entry point for explanation (Tinbergen, 1963).

The core proposal: when a speaker utters a string, two acts are generated simultaneously through different culturally constituted interpretive lenses. We define *inter-lens meaning* (mΔ) as the divergence between these acts:

$$m\Delta \equiv divergence(act_A,\ act_B)$$

$$where\ act_i = f(string,\ lens_i,\ context)$$

When lenses align, $m\Delta \to 0$ and meaning is transparent. When lenses diverge, meaning becomes visible as the gap between speaker action and listener experience. This paper investigates this gap—the between-lenses residual that drives miscommunication, cultural adaptation, and lens co-evolution. Intra-lens meaning (when $m\Delta \approx 0$) is well described by existing frameworks (Grice, 1975; Sperber & Wilson, 1986). The present paper addresses what they cannot.

1.1 Core Terms

We operationalize meaning as an interactional observable, not a metaphysical claim.

Act (act_A, act_B): An interactional state transition triggered by the utterance as interpreted through a lens. Acts are drawn from a finite category set grounded in conversation-analytic primitives (Schegloff, 2007; Stivers et al., 2009): PRAISE, REJECT, TERMINATE, COMMIT, THREATEN, SOFTEN, INVITE, and NEUTRAL. These correspond to recurrent interactional moves documented in conversation analysis. The specific set is a working inventory subject to empirical refinement; the framework requires only that acts be discrete, observable, and cross-culturally applicable. A labeling protocol specifying definitions, exemplars, and decision criteria is provided in Appendix A.

Lens (lens_A, lens_B): A culturally constituted transformation function that maps (string × context) → act. A lens encodes indirect-expression decoding rules, polarity inversion patterns, and plausible deniability expectations. Lenses are operationally measurable via hedge frequency (Hyland, 1998), downgrader/upgrader ratio (Meyer, 2014), politeness marker density (Brown & Levinson, 1987), and irony detection rate (Bryant & Tree, 2005). See Section 5.2.

Context: The situational field (power differential, social distance, formality level, interaction history) shared across act_A and act_B—the same "stage" on which both performances occur. Meaning divergence arises from different lenses applied to the same context.

1.2 Paper Structure

Section 2 documents convergent evolution evidence across eight cultural systems in five unrelated language families. Section 3 extends Pinker et al.'s (2008) evolutionary account with a three-pressure selection model. Section 4 develops the action-theoretic framework, including retroactive meaning determination and the CRLM. Section 5 positions prior theories as special cases and presents testable predictions. Section 6 discusses implications. Section 7 acknowledges limitations and concludes.

2. Convergent Evolution of Indirect Speech

Eight cultural systems spanning five genealogically unrelated language families have evolved structurally isomorphic indirect communication patterns. In each system, the speaker behaves sincerely and protocol-correctly across all observable channels, yet meaning diverges depending on the listener's lens. We first document the English-family cluster, then three genealogically independent systems, and finally test the structural isomorphism statistically.

2.1 English-Family Cluster

Four systems within the English-speaking world share historical contact but exhibit distinct indirect speech patterns that evolved under culture-specific conditions.

British Politeness

Rottier, Ripmeester, and Bush (2010) published an Anglo-Dutch Translation Guide in *Pediatric Pulmonology* because British indirectness was causing professional harm in medical collaboration. The guide documents systematic mappings between surface utterances and intended acts: "I hear what you say" maps to termination of discussion (not acceptance); "With the greatest respect" maps to "You are mistaken" (not deference); "Very interesting" maps to dismissal (not praise); "That is a very brave proposal" maps to "You are reckless" (not admiration); "Quite good" maps to "disappointing" (not positive evaluation); and "I'll bear it in mind" maps to "already forgotten" (not future consideration). In each case, the speaker's tone, demeanor, and conversational behavior are fully consistent with the surface reading—full-channel alignment.

Ripmeester's account of using "very interesting" in an international meeting provides primary-source documentation: Dutch and German colleagues were confused, Australians puzzled, Canadians suspicious, and only British colleagues decoded the intended irony. Same string, same speaker, same context; meaning varied entirely by listener lens (Rottier et al., 2010). Meyer (2014) positions British English at 7.5 on a 10-point indirect negative feedback

scale, and documents the Klopfer case (Section 3.2) as paradigmatic cross-cultural lens failure. Fox (2004) provides broad ethnographic documentation of the pattern.

Corporate American

American corporate communication evolved an independent system: “Per my last email” as implicit reprimand, “Thanks in advance” as obligation encoding. Vine (2004) documents how directives in professional New Zealand English use indirectness to manage power differentials, a pattern replicated in American corporate contexts. Koester (2006) demonstrates that workplace discourse systematically employs indirect speech acts for face management in institutional settings.

Southern US Hospitality

Southern American English provides a regionally distinct system. “Bless your heart” functions as condescension; “Well, that’s a brave choice” encodes criticism. Reed (1993) documents the cultural foundations of Southern indirectness in the broader framework of Southern identity construction. Davies (2015) provides sociolinguistic analysis of Southern politeness markers.

Australian English

Australian English contributes a partial variant: “Yeah nah” encodes refusal, “nah yeah” agreement. Goddard (2012) analyzes Australian communicative patterns in a cross-cultural framework comparing Australian, American, and British interaction styles. The lexical contradiction in “yeah nah” is resolved by lens rather than tone, representing a less formalized indirect system.

2.2 Genealogically Independent Systems

The English-family cluster, while internally diverse, shares historical contact. To establish convergent evolution, we require systems from unrelated language families. Three such systems are documented below.

Japanese Tatemae/Honne (Japonic)

Japanese possesses a meta-linguistic framework (*tatemae/honne*) that explicitly names the surface–depth distinction. The Kyoto dialect is an extreme implementation: 「ぶぶ漬けでもどうどす？」 (offering rice-tea) functions as a departure request; 「考えておきます」 ("I'll think about it") functions as refusal. All operate with full-channel alignment and high deniability. The Japanese system is uniquely metalinguistically aware; other systems operate without explicit naming (Fox, 2004).

Korean Nunchi (한국어, Koreanic)

Korean *nunchi* (눈치, literally "eye-measure") denotes the ability to decode indirect communication through contextual reading (Robinson, 1996; Choi & Choi, 1992). Choi and Choi (1992) identify two key features: nunchi is oriented toward covert, implicit, or indirect communicative exchanges, and nunchi interactions include both "figuring-out" indirect communication and "executing" acts of indirect communication. In Korean interaction, nodding does not indicate agreement but attentive listening—a pattern that frequently causes cross-cultural misunderstanding (foreigners interpret nodding as approval). Direct refusal is systematically avoided in favor of contextual signals, and skilled nunchi is considered a social virtue essential for professional success. Robinson (1996) characterizes nunchi as encompassing tact, savoir faire, and an eye for social situations.

The Korean indirect system is genealogically independent of both the English-family cluster and Japanese. Although Korean and Japanese share typological features (SOV word order, honorific systems), they belong to separate language families with no demonstrated genetic relationship. Chen and Wang (2021) demonstrate systematic cross-cultural differences between Korean and Chinese speakers in first- and second-order indirectness: Korean participants showed different assessments of indirectness and reversed patterns compared to Chinese speakers. This confirms that Korean indirect speech constitutes a distinct cultural lens system not reducible to a generic "East Asian" pattern. The Korean system satisfies the

convergent evolution criterion: it exhibits the {+Surface, −Depth, Full-Channel, High Deniability} signature while emerging from an independent linguistic and cultural lineage.

Persian Taarof (Indo-Iranian)

Persian *taarof* (تعارف) is a ritualized politeness system emphasizing deference, modesty, and indirectness (Koutlaki, 2002; Beeman, 1986). Taarof governs offers, refusals, and invitations through elaborate indirection: a sincere offer and a taarof offer are surface-identical, decodable only through cultural lens. An offer of food may be a genuine invitation or a social ritual requiring polite refusal; the correct response depends entirely on whether the listener applies the taarof lens. Miller, Strong, Vinson, and Brugman (2014) characterize taarof as "ritualized indirectness" in a University of Maryland Center for Advanced Study of Language report, identifying it as a comprehensive interpersonal management strategy encompassing verbal formulas, behavioral scripts, and contextual expectations.

The Persian system is particularly instructive because taarof operates bidirectionally: both speaker and listener engage in coordinated indirectness, with multiple rounds of offer-refusal-insistence constituting a single communicative exchange. This multi-turn structure goes beyond the single-utterance analysis typical of British or Japanese indirect speech, demonstrating that lens-mediated communication can involve temporal extension across turns. Gohari Sadr, Megerdoomian, Emami, and colleagues (2025) introduced TaarofBench, demonstrating that large language models achieve only 40% accuracy on taarof-expected scenarios versus 82% for native speakers—a direct quantification of the lens gap. That performance improved by up to 30 percentage points when prompts were given in Persian rather than English suggests that language form itself partially encodes cultural lens information.[1]

[1] We note that benchmark results of this kind may be sensitive to system-level prompt design and instruction framing, factors that could constitute confounding variables in evaluating cultural competence per se. A systematic analysis of prompt architecture as a lens-setting mechanism falls outside the scope of the present paper.

Akan Verbal Indirection (Niger-Congo)

Akan (Ghana) employs systematic verbal indirection through proverbs, circumlocution, innuendo, metaphors, and euphemisms for conversational management and face maintenance (Obeng, 1994). Obeng (1997) documents how Akan political discourse uses indirection to manage face-threatening acts, with structures paralleling the {+Surface, −Depth} pattern documented in other systems. Nwoye (1992) provides a broader analysis of linguistic politeness in Igbo, another Niger-Congo language, confirming that the pattern extends beyond Akan within the language family.

2.3 Structural Isomorphism and Statistical Confirmation

To move beyond impressionistic comparison, we encode all systems using a uniform template (Table 1). The English-family cluster is treated as a single evolutionary lineage for the convergent evolution test, yielding five independent lineages total.

Table 1

Structural Isomorphism Encoding Across Five Independent Lineages

System	Family	Surface	Depth	Channel	Deny.	Key Source
English-family	Germanic	+ (praise)	− (dismiss)	Yes	High	Rottier et al. 2010; Meyer 2014
Japanese	Japonic	+ (hospit.)	− (termin.)	Yes	High	tatemae/honne lit.
Korean	Koreanic	+ (attend.)	− (refusal)	Yes	High	Robinson 1996; Chen & Wang 2021
Persian	Indo-Iran.	+ (defer.)	− (refusal)	Yes	High	Koutlaki 2002; Beeman 1986
Akan	Niger-Congo	+ (provrb.)	− (critic.)	Yes	High	Obeng 1994, 1997

Five of five independent lineages share the pattern {+Surface, −Depth, Full-Channel Alignment, High Deniability}. A Fisher's method combination of per-parameter binomial tests (each testing whether the observed agreement exceeds chance levels given the parameter's option space) yields a combined $p = .001$ (Monte Carlo verification: $P \approx .00003$, $N = 100{,}000$ simulations; reproducible code and full analysis available at

https://anonymous.4open.science/r/MeaningBetweenLenses-4C64/). Act distributions are probabilistic, not deterministic: in a British–British interaction, P(REJECT | “Very interesting”) ≈ .7; in a British–American interaction, P(PRAISE | “Very interesting”) ≈ .9. Divergence emerges from distributions, inoculating against the “but sometimes it IS literal” objection.

3. Three-Pressure Selection Model

Pinker, Nowak, and Lee (2008) explain why speakers use indirect speech (plausible deniability) but not why the same expression produces different meanings in different listeners, how decoding lenses co-evolve with expressions, or why culturally distinct lens systems emerge. We extend their single-pressure model to three pressures:

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selection_pressure = face_preservation + cooperation_maintenance +
                     hierarchical_ambiguity
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Face preservation (Brown & Levinson, 1987) protects social identity. Cooperation maintenance (Pinker et al., 2008) enables rejection while preserving relationships. Hierarchical ambiguity (Hofstede, 1980) maintains plausible deniability across power differentials. Pinker et al. cover pressure 2 only; the three-pressure model explains why all cultures develop indirectness (pressure 2 is universal), different cultures develop different systems (pressures 1 and 3 vary), and the degree of indirectness varies (the sum varies).

3.1 Empirical Support: PDI × Indirectness

We tested the prediction that PDI correlates with indirect speech using Hofstede’s (2010) publicly available scores and Meyer’s (2014) indirect negative feedback scale. Table 2 presents the data.

Table 2

PDI Scores and Meyer Indirectness Ratings for 13 Cultures

Culture	PDI	Meyer Score	Feedback Style
Israel	13	1.0	Very direct

Netherlands	38	2.0	Direct
Germany	35	2.5	Direct
Denmark	18	3.0	Direct
Austria	11	3.0	Direct
Australia	38	3.5	Moderate
France	68	4.0	Moderate*
USA	40	4.5	Moderate
India	77	7.0	Indirect
UK	35	7.5	Indirect
South Korea	60	8.0	Indirect
China	80	8.5	Indirect
Japan	54	9.0	Very indirect

*Note. PDI from Hofstede (2010). Meyer scores from Meyer (2014). *France is a documented outlier (high PDI, direct feedback).*

Across 13 cultures, PDI correlates positively with Meyer's indirectness (Spearman $\rho = .69$, $p = .009$; Pearson $r = .70$, $p = .008$). The result is robust to exclusion of France ($\rho = .72$, $p = .008$). Indirect cultures (Meyer > 5.0) have significantly higher PDI than direct cultures (Meyer < 4.0): mean 61.2 vs. 25.5, Mann–Whitney $U = 27.5$, $p = .014$. These results constitute an exploratory analysis ($N = 13$) and require confirmation with larger-scale cross-linguistic corpora. However, the effect size ($\rho = .69$, medium-to-large) is sufficient to support the directional prediction.

3.2 The UK Anomaly

The UK (PDI = 35) scores as highly indirect (Meyer = 7.5) while Germany (PDI = 35) scores as highly direct (2.5)—an indirectness gap of 5.0 despite identical PDI. This dissociation is what the three-pressure model predicts: PDI captures hierarchical ambiguity (pressure 3) only. British indirectness is driven by face preservation (pressure 1), operating independently. The Klopfer case (Meyer, 2014)—a German finance director who misinterpreted months of British downgraded directives—instantiates this gap: both cultures share a PDI stratum, yet meaning divergence was maximal.

3.3 Expression–Lens Co-evolution

The convergent evolution documented in Section 2, combined with the three-pressure selection model, suggests a co-evolutionary dynamic between indirect expressions and decoding lenses.

Social pressure (cooperation maintenance + rejection necessity) drives the initial evolution of indirect expressions (Pinker et al., 2008: plausible deniability). Within groups, agents who correctly decode indirect acts gain social navigation advantage—they can distinguish genuine offers from ritual ones, optional suggestions from mandatory directives, and literal praise from ironic dismissal. Successful decoders' lenses are copied through social learning, leading to within-group convergence on shared expression×lens systems. This within-group convergence is observable in established systems: Kyoto dialect users share a stable decoding lens for *bubu-zuke*; British English speakers share stable decodings for "very interesting"; Persian speakers share stable taarof protocols.

Between groups, weak inter-group contact prevents lens transfer. Each group's expression×lens pair drifts independently, producing different surface expressions for structurally equivalent functions. At group boundaries, systematic divergence is maximized: agent_i uses expressions from group 1 while agent_j applies the lens of group 2. This boundary effect *is* the "miscommunication as observation condition" central to this paper. Within each group, expression complexity and lens precision co-evolve in a spiral—more nuanced expressions require more precise lenses, which in turn enable even more nuanced expressions.

The co-evolutionary model naturally produces the observed pattern: universal indirect speech (all groups evolve it independently under pressure 2), structural isomorphism (the same three pressures produce the same architectural signature), cultural specificity (different expressions and lenses per group), and boundary divergence (maximum misunderstanding where systems meet). Full agent-based simulation is reserved for future work; the model is presented as a conceptual framework whose qualitative predictions match observed data.

The analogy to eye evolution is instructive but incomplete. Same selection pressure (light detection / polite rejection), independent evolution, different structures (retinal orientation / different expressions), equivalent function (vision / plausible deniability)—but the language case has an additional dimension: each indirect speech system co-evolves a *decoding apparatus* (the cultural lens) unique to itself. The octopus eye and mammalian eye both detect light directly; the Kyoto system and the British system each require a culturally specific decoder. This co-evolved decoding apparatus—absent in the biological analogy—is what generates the meaning divergence documented in this paper.

4. The Lens: From Ontology to Action Theory

4.1 Departures from Prior Frameworks

Austin (1962) assumes one utterance maps to one illocutionary act. We revise: the illocutionary act itself is lens-dependent. “Very interesting” is simultaneously REJECT (British lens) and PRAISE (American lens). The act is not invariant across lenses.

Grice (1975) assumes maxim violations are detectable. We show detection is lens-dependent: without the British lens, nothing signals violation in “Very interesting.” Without detection, inference cannot begin. Gricean implicature is thus a special case where lens_A ≈ lens_B. Additionally, Japanese *baka-teinei* (excessively polite sarcasm) does not violate maxims but *over-complies*. We propose this as a fifth category of maxim exploitation: *hyper-compliance*. Where flouting generates meaning by visibly breaking a maxim, hyper-compliance generates meaning by visibly exceeding it—preserving full deniability while conveying frustration to lens-equipped listeners.

Sperber and Wilson’s (1986) Relevance Theory proposes that “manifestness” determines processing. We show manifestness is culturally constituted: same cognitive capacity + different lens → different optimal relevance. Relevance Theory describes intra-lens processing; this paper describes inter-lens divergence.

4.2 Meaning Determination Under Full-Channel Alignment

Under full-channel alignment—when all observable channels (linguistic, paralinguistic, non-verbal, contextual) are consistent with the surface reading—mΔ is typically underdetermined for out-group listeners. The divergence is not detectable by increasing observational resolution, because signal detection itself is lens-dependent. This underdetermination is specific to inter-lens meaning (mΔ); intra-lens meaning remains fully determined within any given lens.

4.3 Retroactive Meaning Determination

Meaning is not fixed at utterance. Post-hoc lens changes retroactively re-determine past utterances via three operators:

Operator 1—Lens Update: $\text{lens_B}(t_1) = \text{UPDATE}(\text{lens_B}(t_0), \text{corrective_event})$. The listener's lens is modified by a high-cost event (crisis, reprimand, cultural instruction, or social correction).

Operator 2—Retro-decode: $\text{act_B}(t_1) = f(\text{string}, \text{lens_B}(t_1), \text{context})$. The same string, received at t_0, is re-decoded through the updated lens. The string has not changed; the act has.

Operator 3—Time-dependent mΔ: $m\Delta(t_0) = 0$ (divergence undetected) $\rightarrow$ $m\Delta(t_1) \gg 0$ (divergence emerges retroactively). Prior utterances retroactively become commands, dismissals, or threats depending on the updated lens.

The Klopfer case (Meyer, 2014) is the standard-form paradigm. For months, Klopfer received downgraded directives from his British manager ("perhaps you should think about…") and decoded them through his German (direct) lens as optional suggestions. At the crisis point (a reprimand), his lens was updated: he acquired the decoding rule "strip the downgrader, extract the core act." At this point, all prior utterances retroactively became mandatory orders—the same strings, the same contexts, but different acts. The time scale (months) demonstrates that retroactive meaning determination operates across extended periods. The Charleston "bless your heart" case (3-hour delay) and the *bubu-zuke* case (variable

delay) share identical operator structure at different time scales. Three operators are sufficient to describe all documented cases.

4.4 The Cultural–Relational Lens Matrix (CRLM)

Lenses have at least two independent sources: cultural (group-level, acquired through socialization) and relational (dyad-level, co-constructed between individuals). We extend acts to two layers: act = (act_surface, act_deep).

Table 3

CRLM: Four Decoding Conditions with Exemplars

		Relational Absent (R_0)	**Relational Present (R_1)**
Cultural	**Absent (C_0)**	LITERAL: Tourist eats bubu-zuke; stranger hears “idiot” as insult	RELATIONAL: Partner reads “idiot” as affection (dyad code)
Lens	**Present (C_1)**	CULTURAL: Kyoto native reads bubu-zuke as “leave”; recognizes taarof pattern	RESONANT: Intimate partner in shared culture; deepest alignment

The CRLM resolves the mΔ = 0 quality problem: Literal and Resonant decoding both produce mΔ = 0, but at different depth levels. Literal alignment is shallow transparency; resonant alignment is deep resonance.

Mirror-Image Inversions

Kyoto-type inversion (+Surface → −Depth) operates through cultural lenses and exhibits convergent evolution (Section 2). Tsundere-type inversion (−Surface → +Depth) operates through relational lenses: surface-negative expressions encode deep-positive meaning, independently attested in Japanese, French, Spanish, and Korean. That the same three pressures produce polarity-inverted patterns on different substrates demonstrates the model’s generality.

Error Modes: Underdecode and Overdecode

Underdecode: lens insufficient → depth missed → listener falls to surface reading → *invisible mΔ* (divergence exists but is undetectable). Examples: tourist eating bubu-zuke; Klopfer reading “suggestions” literally; foreign colleague missing taarof signals.

Overdecode: lens applied where not warranted → phantom depth generated → *phantom mΔ* (divergence detected but nonexistent). Examples: interpreting genuine criticism as tsundere affection; reading malice into sincere British praise; projecting taarof refusal onto a genuine decline.

Table 4

Predicted Error Modes by CRLM Quadrant Mismatch

Actual Condition	Listener Assumes	Error	mΔ Type	Example
Cultural (C_1)	Literal (C_0)	Underdecode	Invisible	Tourist/bubu-zuke
Literal (C_0)	Cultural (C_1)	Overdecode	Phantom	Malice in praise
Relational (R_1)	Literal (R_0)	Underdecode	Invisible	Outsider/inside joke
Literal (R_0)	Relational (R_1)	Overdecode	Phantom	Projected affection

Relational Lens Legitimacy

Cultural lenses can be held unilaterally; relational lenses must be co-constructed, bidirectionally recognized, and revocable by either party. When these conditions are not met, the lens is projection, and Overdecode errors produce phantom mΔ.

4.5 The Bubu-zuke Proof

The *bubu-zuke* case provides a three-pattern demonstration that separates intention from meaning and validates the action-theoretic framework:

Pattern 1—Tourist (no lens): act_A = Offer tea (surface). act_B = Accept tea. mΔ = 0. The meaning "please leave" is *not generated* in the interaction space. The speaker intended it, but meaning does not exist without the inter-lens divergence. This demonstrates the fundamental separation of intention from meaning: intention drives act_A but does not constitute meaning.

Pattern 2—Kyoto native (lens present): act_A = Offer tea (surface, identical to Pattern 1). act_B = "I should leave" (decoded through Kyoto lens). mΔ ≠ 0. The meaning "please leave" emerges—not from the speaker's mouth, not from the listener's imagination, but

from the divergence between lens-mediated acts. The utterance itself is identical across patterns; the meaning differs because the lens differs.

Pattern 3—Tourist learns later (retroactive determination): At t=0: act_B = Accept tea, mΔ = 0 (no inter-lens meaning). At t=1: the tourist's friend explains the cultural code (lens update). At t=2: the tourist retro-decodes—"That was a dismissal!"—and mΔ shifts from 0 to a positive value. The same three-operator structure as the Klopfer case (Section 4.3), demonstrating that retroactive meaning determination operates at scales from hours to months.

These three patterns yield three implications. First, meaning ("please leave") was never "hidden inside" the utterance; it was generated by divergence. Second, when mΔ = 0 (tourist case), the inter-lens meaning literally does not exist, though intra-lens meaning ("tea offered") remains. Third, the same utterance can gain meaning retroactively through lens acquisition. Together, these establish the separation of intention from meaning that is central to the framework.

5. General Framework

5.1 Prior Theories as Special Cases

Each prior framework is recovered as a special case under specific boundary conditions. *Grice is recovered when* lens_A ≈ lens_B: shared cultural framework enables maxim violation detection, implicature recovery proceeds as Grice describes, and mΔ ≈ 0. Gricean pragmatics is the theory of what happens when lenses are already aligned. *Sperber and Wilson is recovered when* analysis remains within a single lens: cognitive environment is lens-internal processing, relevance optimization describes intra-lens meaning (*a*), not mΔ. *Pinker et al. is recovered when* the speaker correctly estimates the listener's lens distribution: plausible deniability becomes intentional mΔ management, and the game-theoretic analysis describes the speaker's strategic calculus.

The CRLM makes a distinctive prediction not available to these frameworks: it predicts not only *when* indirect meaning is decoded, but *how* decoding fails. Underdecode yields invisible divergence (miscommunication without awareness)—a state where both parties believe communication has succeeded. Overdecode yields phantom divergence (perceived offense without intent)—a state where the listener detects hostility that was never encoded. This distinction between invisible and phantom mΔ is not captured by standard implicature models, which assume that failure to decode is detectable as a processing difficulty.

5.2 Lens Operationalization

Lenses are operationalizable as vectors with four measurable components: hedge frequency (hedges per 100 words; Hyland, 1998), downgrader ratio (ratio of downgraders to upgraders in negative feedback contexts; Meyer, 2014), politeness marker density (frequency of positive/negative politeness strategies per interaction unit; Brown & Levinson, 1987), and irony detection rate (proportion of ironic utterances correctly identified in forced-choice tasks; Bryant & Tree, 2005).

Existing instruments enable measurement. Meyer's (2014) upgrader–downgrader axis provides a scalar proxy for lens directness. Brown and Levinson's (1987) politeness strategies serve as lens components. Hofstede's (1980) PDI functions as a macro-level correlate. Crucially, the Anglo-Dutch Translation Guide (Rottier et al., 2010) is, unknowingly, a direct empirical measurement of mΔ: its three-column structure (string, British act, Dutch interpretation) records act_A and act_B per expression. If this three-column format can be independently generated for other culture pairs (Japanese–American, Korean–German, Persian–Australian), the resulting collection constitutes a cross-cultural lens measurement dataset—convergent evolution of the measurement instrument itself.

5.3 Divergence Quantification

We propose Jensen–Shannon Divergence (JSD) as the primary mΔ metric. JSD is preferred over KL divergence because it is symmetric (divergence(A,B) = divergence(B,A)) and

bounded ($0 \leq JSD \leq 1$), matching the intuition that meaning divergence is mutual. For a given string S and cultural groups A and B, $P_A = P(Act \mid S, lens_A)$ and $P_B = P(Act \mid S, lens_B)$, and $m\Delta(S) = JSD(P_A, P_B)$.

A minimal pilot implementation would require 10–20 exemplar expressions from the documented cultural systems (e.g., “Very interesting,” “Per my last email,” “Bless your heart,” *bubu-zuke*, “Yeah nah”), presentation with full context to participants from 2+ cultural groups ($N \geq 30$ per group), forced-choice act label assignment from the Definition Box categories (Appendix A), JSD computation per expression, and correlation of JSD scores with documented miscommunication rates. Expressions documented as miscommunication-prone should show high JSD; cross-culturally transparent expressions should show $JSD \approx 0$. This design would transform the paper’s theoretical framework into a testable measurement model.

5.4 Testable Predictions

The framework generates five specific, falsifiable predictions testable with existing datasets and standard experimental paradigms:

Prediction 1 (Lens–Divergence): Hedging frequency × cultural distance predicts interpretation divergence. Cultures with higher downgrader ratios show greater interpretation divergence when their utterances are evaluated by upgrader cultures.

Prediction 2 (Hierarchy–Indirectness): Hofstede PDI correlates positively with indirect speech frequency. Preliminary evidence is reported in Section 3.1 ($\rho = .69$, $p = .009$, $N = 13$, exploratory).

Prediction 3 (Retroactive Determination): When participants are exposed to culturally coded indirect utterances and subsequently provided with cultural context (lens update), their retrospective interpretation of the same utterance shifts significantly. Testable via standard pre/post experimental design with cultural priming.

Prediction 4 (Convergent Structure): Cultures that independently developed indirect speech systems show structural isomorphism in their expression–meaning mapping patterns. Evidence: five genealogically independent lineages (Section 2.3, Fisher's $p = .001$).

Prediction 5 (Emotion–Indirectness Link): Emotion tone as measured by articulatory features correlates with indirect speech frequency, as both are driven by the same body-state → sound → string pipeline. This bridges the present framework with [Author] (submitted) and constitutes a cross-paper test of the integrated theory stack.

6. Discussion

6.1 Implications for Translation

Traditional translation assumes meaning is "in" the source text and must be "moved" to the target. The lens framework suggests translation is lens transformation: reconstructing conditions under which equivalent divergence patterns arise in the target culture. A translator working with *bubu-zuke* does not transfer a "hidden meaning" but must reconstruct a surface–depth pattern that activates the equivalent lens in the target culture. A literal translation ("Would you like some rice tea?") preserves the string but destroys the divergence pattern, because English lacks a cultural lens that maps tea offerings to departure requests. A functional translation ("Well, it's getting late, isn't it?") reconstructs the mΔ pattern in an English cultural key. The quality of translation is fidelity to the mΔ pattern, not to the string.

Similarly, translating Persian taarof into English requires not literal rendering of polite formulas but reconstruction of the multi-turn offer-refusal-insistence dynamic in culturally appropriate English equivalents. The lens framework provides a precise criterion for evaluating translation quality: does the translation preserve the act distributions that the original generates across lens-equipped and lens-unequipped listeners?

6.2 Implications for NLP

Current NLP systems implicitly average across lenses rather than maintaining lens-conditional distributions. When processing “Very interesting,” the model outputs a blended probability distribution over acts rather than lens-conditional distributions. The CRLM framework suggests that cultural lens parameters should be explicit inputs to interpretation models, enabling the same model to generate different act distributions under different lens settings.

The TaarofBench result (Gohari Sadr et al., 2025)—40% LLM accuracy versus 82% native speakers—directly quantifies the lens gap in current systems. That performance improved 30 percentage points when prompts shifted from English to Persian suggests language form itself partially encodes lens information—a finding consistent with the present framework’s prediction that lenses are embodied in linguistic structure, not just in abstract cultural knowledge. This has practical implications: culturally sensitive NLP may require not just cultural metadata but language-form-mediated lens activation.

6.3 Miscommunication as Observation Condition

Cross-cultural miscommunication is not communicative failure but the condition under which inter-lens meaning becomes observable. “Successful” communication ($m\Delta \approx 0$) is the special case requiring lens alignment, not the default. This reframing shifts the research object: instead of asking “How does communication succeed?” (which presupposes alignment), we ask “How does divergence become invisible?” (which reveals the role of shared lenses in producing the illusion of transparent meaning).

The clinical context underscores this. Rottier et al. (2010) published in a medical journal because invisible divergence—both parties believing communication had succeeded while holding mutually inconsistent interpretations—was causing real professional harm. The most dangerous miscommunication is the kind neither party detects: high mutual confidence in mutually inconsistent interpretations. The framework predicts this occurs maximally when surface-level communication is fluent (no signal of difficulty) and lens divergence is large

(different act distributions). The Klopfer case is paradigmatic: for months, both Klopfer and his manager believed their communication was working.

6.4 Within-Language Lens Inversion: The Aho/Baka Case

The framework predicts that lens effects should operate at finer granularity than national culture. A striking within-language demonstration is provided by the Japanese regional terms *aho* and *baka*, both dictionary synonyms meaning "foolish." In Osaka (Kansai), *aho* functions as an affection marker ("you lovable fool") while *baka* constitutes a genuine insult. In Tokyo (Kanto), the polarity reverses completely: *baka* functions as an affection marker while *aho* constitutes an insult. Same language, same dictionary entry, same speech act category—only the regional lens differs.

This case is theoretically significant for three reasons. First, it eliminates confounds present in cross-cultural cases: language difference, cultural distance, and translation ambiguity are all absent. The lens effect is isolated in its purest form. Second, the inversion is symmetric and exhaustive: the two regions do not merely differ in degree of offense but assign opposite affective polarities to the same pair of synonyms. This constitutes a third type of mirror-image inversion beyond the Kyoto-type (+Surface → −Depth, cultural) and tsundere-type (−Surface → +Depth, relational): a *regional polarity swap* in which the depth layer's sign reverses across regional lenses applied to dictionary-identical items. Third, this pattern is inexplicable under standard Gricean analysis, because no maxim violation occurs in either region—*aho* as affection in Osaka is not a flouting or implicature but a conventional regional meaning encoded in the Osaka lens. Relevance Theory likewise cannot account for the reversal, since the cognitive environments of Osaka and Tokyo speakers are comparable; only the lens differs.

6.5 Relational Dimension and Interpretive Consent

The CRLM demonstrates that lenses have at least two independent sources. This paper focuses primarily on cultural lenses, where convergent evolution evidence is strongest and cross-

cultural data is available. However, the relational dimension opens a significant research program.

The framework predicts that communicative intimacy can be formally defined as the depth at which $m\Delta = 0$ is achieved: literal alignment (surface only) versus resonant alignment (both layers). A couple who share both cultural context and relational codes operate in the Resonant quadrant of the CRLM; their communication achieves depth-level alignment that is qualitatively richer than the surface-level alignment between strangers, even though both register as $m\Delta = 0$.

We name the pre-behavioral layer of communicative alignment *interpretive consent*—shared agreement on how an utterance is to be decoded. Interpretive consent spans the full range of lens types: culturally shared indirect speech patterns (*bubu-zuke*, “Very interesting”, taarof protocols), professionally shared codes (technical jargon decoded within a discipline but opaque outside it), small-group codes (in-group catchphrases, family shorthands), and relationally co-constructed affective codes. The term is used in its analytical sense (mutual agreement on interpretation), not in any domain-specific normative sense. What unifies these cases is that decoding requires shared participation in the code’s construction or adoption; absent this participation, the code is inert and depth remains inaccessible. Fuller treatment is reserved for dedicated future work.

6.6 Theory Stack: Integration with Prior Work

The present framework constitutes the upper layer of a meaning-generation pipeline initiated in [Author] (submitted). That paper describes how involuntary physiological reflexes leave articulatory traces in sound—the body → sound layer, cross-validated across large-scale typological databases. The present paper extends this pipeline upward: sound → string → lens interaction → meaning divergence.

Together, the two papers propose a complete pipeline from body to meaning with no arbitrariness at either end. [Author] (submitted) establishes that the material of language

(sound) is grounded in bodily reflexes. The present paper establishes that the interpretation of language (meaning) is emergent from lens-mediated divergence. The integration generates Prediction 5: if both layers share the same body-state → expression pipeline, then emotion-laden sound patterns (documented in [Author], submitted) should correlate with indirect speech frequency, as both are downstream effects of articulatory-affective grounding. This cross-paper prediction provides both a bridge between submissions and a test of the integrated theory stack.

6.7 Observer-Dependent Determination

The structure of retroactive meaning determination—in which the interpretive framework applied to an already-recorded event determines its meaning post hoc—resonates with observer-dependent determination structures found across cognitive science and information theory. In both cases, the observed quantity (meaning / measurement outcome) is not intrinsic to the observed system but is co-constituted by the observer's interpretive apparatus. This structural parallel suggests that meaning generation may instantiate a more general epistemological principle regarding the constitutive role of interpretive frameworks in the determination of experiential content. A fuller exploration of this parallel is reserved for future work in a philosophical venue.

7. Limitations and Conclusion

7.1 Limitations

Several limitations warrant acknowledgment. First, the convergent evolution evidence draws on five independent lineages (eight systems). Extension to additional lineages—Middle Eastern indirect refusal beyond taarof, Igbo politeness (Nwoye, 1992), Mopan Maya intent-free communication, Ethiopian indirect speech patterns—would further strengthen the convergent evolution argument. Second, the PDI × indirectness correlation is exploratory (N = 13) using Meyer's qualitative scale as a proxy for indirectness. Larger-scale cross-linguistic corpus analysis with direct indirectness measures (hedge frequency, downgrader ratio) is needed; the

effect size (ρ = .69) is sufficient to support the directional prediction, but confirmation with independent data is required.

Third, the act categories (Appendix A) are a working set grounded in conversation analysis. Empirical refinement through cross-cultural forced-choice tasks will likely yield a more precise and complete category set; inter-annotator agreement (κ) should be reported in any application. Fourth, the co-evolution model (Section 3.3) is conceptual rather than computational; agent-based simulation would provide stronger demonstration of the model's sufficiency. Fifth, the CRLM's relational lens dimension is introduced but not empirically tested; this dimension requires dedicated experimental work with dyad-level data. Sixth, the English-family cluster shares historical contact; while we treat it as a single lineage for the convergent evolution test, future work should investigate whether the within-family variants (British, Corporate, Southern, Australian) constitute independent emergences or historical diffusion.

7.2 Conclusion

This paper has reframed the central question of meaning from "Where is meaning?" to "What is meaning doing?" We have proposed that meaning is the divergence between the act a speaker attempts and the act a listener experiences, generated through the interaction of culturally constituted interpretive lenses. Eight cultural systems across five genealogically unrelated language families—English-family (Germanic), Japanese (Japonic), Korean (Koreanic), Persian (Indo-Iranian), and Akan (Niger-Congo)—provide convergent evolution evidence for structural isomorphism in indirect communication patterns, statistically confirmed at $p = .001$. A three-pressure selection model (face preservation + cooperation maintenance + hierarchical ambiguity) extends Pinker et al.'s (2008) single-pressure account and correctly predicts both the universality and cultural variation of indirect speech, with preliminary PDI × indirectness correlation (ρ = .69, $p = .009$) supporting the hierarchical ambiguity component.

The Cultural–Relational Lens Matrix (CRLM) identifies two independent lens sources, four decoding conditions, mirror-image inversion patterns, and two systematic error modes

(Underdecode/Overdecode). The framework recovers Gricean implicature, Relevance Theory, and strategic indirect speech as special cases under specific boundary conditions, introduces hyper-compliance as a fifth category of Gricean maxim exploitation, and generates five testable predictions. We have argued that cross-cultural miscommunication is not communicative failure but the condition under which inter-lens meaning becomes observable—the most theoretically interesting and evolutionarily consequential component of meaning.

The title "Meaning Between Lenses" itself demonstrates the theory: "between" can be read as spatial or action-theoretic. Which reading you get depends on your lens.

Appendix A: Act Category Labeling Protocol

The following protocol specifies definitions, exemplars, and decision criteria for the eight act categories used in this paper. It is designed for use by coders in forced-choice labeling tasks.

Act	Definition	Canonical Example	Decision Rule
PRAISE	Positive evaluation of addressee/topic	"That's excellent work"	Speaker attributes positive value
REJECT	Negative evaluation or dismissal	"I don't think so"	Speaker denies, dismisses, or criticizes
TERMINATE	Signal to end interaction	"Well, it's getting late"	Speaker moves to close exchange
COMMIT	Binding speaker to future action	"I'll handle it"	Speaker takes on obligation
THREATEN	Imposing cost on addressee	"You might want to reconsider"	Speaker signals negative consequence
SOFTEN	Mitigating face threat	"With respect…"	Speaker preemptively reduces threat
INVITE	Opening or extending interaction	"Would you like to join?"	Speaker creates opportunity for addressee
NEUTRAL	No interactional state change	"The meeting is at 3 PM"	Pure information transfer

Coders should: (1) read the full utterance in context, (2) select the single best-matching act category, (3) record confidence (High/Medium/Low). Ambiguous cases where two categories

compete should be flagged for discussion. Inter-annotator agreement (κ) should be reported for any empirical application of these categories.

Data Availability Statement

All statistical analyses, reproducible code, and supplementary materials for this paper are available in an anonymous repository: https://anonymous.4open.science/r/MeaningBetweenLenses-4C64/. Hofstede's PDI scores are publicly available at geerthofstede.com/research-and-vsm/dimension-data-matrix/. Meyer's (2014) indirectness ratings are derived from *The Culture Map*. No novel empirical data were collected for this study; all analyses use publicly available datasets and published sources.

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